

Time value of money formulas

(1.) Amount of Interest from t_1 to t_2 if $\$K$ is invested: $A_K(t_2) - A_K(t_1)$

(1.) Effective interest rate from t_1 to t_2 : $i_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_1)}$

(2.) Effective discount rate from t_1 to t_2 : $d_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_2)}$

(3.) Simple Interest accumulation function: $a(t) = 1 + rt$ (where $r = i_{[0,1]}$)

(4.) Simple Discount accumulation function: $a(t) = \frac{1}{1 - rt}$ (where $r = d_{[0,1]}$)

(4.) Compound Interest accumulation function: $a(t) = (1 + i)^t$ (where $i = i_{[t, t+1]}$ for any t)

(5.) Discount function: $v(t) = \frac{1}{a(t)}$ and for compound interest $v = v(1) = \frac{1}{1 + i}$

(6.) $1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$ where $i^{(m)}$ is the nominal interest rate compounded m times per year

(7.) $1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m$ where $d^{(m)}$ is the nominal discount rate compounded m times per year

(8.) $d = \frac{i}{1 + i}$ and $i = \frac{d}{1 - d} = \frac{d}{v}$

(9.) Constant force of interest or nominal rate convertible continuously:

$$\delta = \ln(1 + i) \iff e^\delta = 1 + i$$

(10.) Force of interest at time t : $\delta_t = \frac{a'(t)}{a(t)} = \frac{d}{dt} (\ln(a(t))) \iff a(t) = e^{\int_0^t \delta_r dr}$